

Title: *Linguistic modality (hand)shapes your expectations: An EEG study of form-based predictions during naturalistic comprehension of stories in American Sign Language*

Abbreviated Title: Linguistic prediction in American Sign Language narratives

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Authors report no conflict of interest. All pre-registrations, data, code, and study materials are publicly available on Open Science Framework (OSF, [peer review link](#)). Correspondence concerning this article should be addressed to Anthony Yacovone, 33 Kirkland St., Cambridge, MA 02138. Email: anthony.yacovone@tufts.edu

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Abstract

Contemporary theories of linguistic prediction are largely based on our understanding of written and spoken languages. Learning to read has been argued to enhance the prediction not only of written words, but also of spoken words in the same language. This hypothesis suggests that prediction might be very different in a language *without* a primary written form like American Sign Language (ASL). To address this question, we used electroencephalography (EEG) to assess prediction in deaf ASL signers, who are also proficient English readers. Specifically, we asked whether signers predict the form of upcoming words in an ASL story. We factorially manipulated predictable and unpredictable signs to be either the original sign or a nonsensical sign with a similar or dissimilar form. In ASL, signs can be identified by three parameters: *handshape*, *location*, and *movement*. Thus, form-similar non-signs overlapped in location and movement features, whereas form-dissimilar non-signs had no overlap with the original sign. Evidence for prediction would manifest as smaller N400s to predictable relative to unpredictable signs. If these predictions are form-based, we expect smaller N400s to form-similar non-signs in predictable contexts. We found evidence of the first pattern but not the second. Exploratory analyses, however, revealed smaller N400s to non-signs with similar handshapes to the originally predicted signs (disregarding location and movement). This would be expected if signers only predicted upcoming handshapes during comprehension. Further work is necessary, but this study suggests that the modality of a language may shape which form representations are targeted for prediction.

Keywords: sign language, form-based prediction, naturalistic discourse, ERPs, N400.

1. Introduction

Prediction research has long been dominated by studies of written and spoken languages. These two systems for encoding a language are tightly linked—after all, written text was largely created to record things that we might otherwise say (Bloomfield, 1984, Chapter 2; but see, Olson, 1993). Thus, the written and spoken versions of a language make use of the same lexical items, syntactic structures, and semantic representations (e.g. Caramazza, 1997; Cleland & Pickering, 2006). As a result, the experiences that we have in one modality can influence our comprehension and production in the other modality (Braze et al., 2007; Huettig & Pickering, 2019; Monzalvo & Dehaene-Lambertz, 2013; Storch & Whitehurst, 2002).

Recently, Huettig and Pickering (2019) proposed that learning to read enhances the ability to predict upcoming words, not only in the written modality but also in the spoken modality. Specifically, they argue that reading provides optimal conditions for the development of prediction (e.g. reading can unfold in a stable, self-paced manner, and written text is largely invariant in form across contexts). They propose that as reading proficiency improves over time and with experience, people begin to predict which written words are likely to come next in a sentence *before* they encounter them on the page. Once these predictive skills are well-established in reading, they can also be used to predict upcoming spoken words when listening to the same language.

On this account, the ability to predict upcoming words (and their form features) during comprehension is not a central characteristic of the language system but rather a cultural artifact that emerges under particular developmental circumstances. Specifically, the transfer of predictive skills from the written to the spoken modality occurs due to their shared systems of representation. Preliminary support for this claim comes from psycholinguistic experiments showing that higher

reading proficiency in a language is associated with making form-based predictions about upcoming spoken words (and phonological forms) in the same language (Favier et al., 2021; Mani & Huettig, 2012, 2014; Mishra et al., 2012; Ng et al., 2018).

This proposal sparks an important question—what happens when a language does not have a primary written form? For example, visual-manual or sign languages like American Sign Language (ASL) are fully realized, autonomous linguistic systems with semantic, syntactic, and lexical complexity equivalent to spoken languages (Bellugi & Fischer, 1972; Hickok et al., 2001; Stokoe, 1980; for discussion, see Emmorey & Lane, 2013). However, unlike most spoken languages, they do not have a commonly used written form (Grushkin, 2017). As a result, deaf signers who use a visual-manual language as their primary mode of communication often learn to read and write in another language, like English (Hoffmeister & Caldwell-Harris, 2014; Musselman, 2000). The present study explores whether such signers, who do not engage directly with their primary language through written text, still demonstrate form-based prediction when comprehending a narrative in the visual-manual modality.

In the remainder of this Introduction, we first describe the process of learning to read in hearing and deaf signing populations and consider the question of whether predictive skills from reading should transfer to the deaf signers' processing of sign languages. Next, we discuss how researchers typically demonstrate prediction in written and spoken languages, drawing on research that uses electroencephalography (EEG). Then, we review the handful of EEG studies in ASL, noting that these studies do not provide definitive evidence for prediction in the visual-manual modality. Finally, we end by describing the present study and how it is designed to assess prediction in ASL.

1.1. How do hearing and deaf signing populations learn to read?

The process of learning to read is fundamentally different for hearing children and deaf signing children with limited or no hearing ability (Goldin-Meadow & Mayberry, 2001; Lederberg et al., 2012). For hearing children, learning to read largely consists of mapping new written symbols onto their existing phonological and lexical representations. For example, in an alphabetic language like English, hearing readers must learn how graphemes map onto the phonemes of their spoken language (Ehri, 1987; Gibson et al., 1963). After learning these mappings, they can successfully decode written words into the spoken words that they already know. As hearing children become more proficient readers, they begin to build up direct associations between written word forms and their lexical concepts, allowing them to access a word's meaning without sounding it out (Rayner et al., 2001; Seidenberg, 2017).

In contrast, deaf children learning ASL as a primary language are faced with a different task, as there is not a direct correspondence between the written text and their primary language. As a result, the process of learning to read for deaf signers is bilingual: it involves simultaneously learning the written word forms, the lexicalized concepts that they encode, and the syntactic system of a new language (see Goldin-Meadow & Mayberry, 2002). For example, a deaf child learning to sign in ASL *and* read in English will need to acquire the lexicalization patterns and syntactic structures unique to both languages (and modalities). Thus, mapping between a sign language and a written language will involve translating from one language to another rather than merely acquiring a new perceptual input system.

Nevertheless, there is some evidence that deaf readers, like hearing readers, make connections between written words and the corresponding lexical representations in their primary language. For example, Treiman and Hirsh-Pasek (1983) demonstrated that native English

speakers struggled more when silently reading sentences with similar sounding words (e.g. “She chose three shows to see at the theater”) relative to sentences with dissimilar sounding words (e.g. “She picked two movies to see with her friend”). In contrast, native deaf ASL signers did not show difficulties with these sound-similar sentences, but they did experience difficulties when the sentences contained English words that corresponded to signs with similar articulatory features (e.g. “I ate the apples at home yesterday” in which the ASL signs for EAT, APPLE, HOME, and YESTERDAY all involve closed handshapes articulated near the face).

So far, we have discussed the very real challenge of learning to read in a language that you do not already know. It is critical to point out that this challenge does not provide empirical support for a policy of limiting deaf children’s access to sign language. In fact, numerous studies have shown that being *more* proficient in a sign language is associated with stronger reading abilities (Andrew et al., 2014; Chamberlain et al., 1999; Strong & Prinz, 1997; see Goldin-Meadow & Mayberry, 2002). It is clear that the advantages of having an early and full access to a natural language upon which to build further language skills far outweigh any difficulties experienced when acquiring literacy via a second language system.

The present study is motivated by the question of whether prediction in a primary linguistic modality (like speech or sign) requires experience with a written form of that same language. According to Huettig and Pickering (2019), the primary reason that predictive skills transfer from written to spoken language is that these language systems act on shared representations. Thus, the process of reading develops strong associations between particular words and broader linguistic contexts, and those associations are also available during spoken language comprehension of that same language. On this account, predictive skills should transfer from written to spoken modalities

within the *same* language, but not necessarily between *different* languages. For example, the associations learned when reading in English may not necessarily transfer to listening to Japanese.

The difficulty with testing this account is that many users of spoken languages learn to read in their native language, and thus prediction is likely to emerge regardless. For these reasons, studying prediction in sign language is an important next step. As a starting point for addressing these open questions, the present study seeks to understand predictive processing in sign comprehension. In particular, we are interested in whether signers make lexical predictions and anticipate the form of upcoming signs in an ASL narrative. In the sections below, we summarize the prior literature using EEG to demonstrate prediction in written and spoken languages. Then, we describe how EEG has been used in sign language research, noting that there is very little evidence to date that prediction of any kind occurs in the visual-manual modality.

1.2. Predictive processing in written and spoken languages—evidence from EEG

Psycholinguists have long used EEG and event-related potentials (ERPs) to study the nature of prediction during comprehension (Federmeier, 2007; Kuperberg & Jaeger, 2016; Kutas et al., 2006; Kutas & Van Petten, 1994; Swaab et al., 2012). To do this, researchers typically record comprehenders' neural responses to words presented in various ways: in prime-target pairs (e.g. salt–PEPPER), in sentential contexts, or even in larger discourse narratives (Kutas & Hillyard, 1984; Lau et al., 2013; van Berkum, 2004; Yacovone et al., 2021). The responses to each target word are then averaged together, revealing systematic patterns of neural activity known as ERP components (Kappenman & Luck, 2011; Luck, 2014). The most common and best understood ERP component in prediction studies is the N400. This component is a negative-going deflection

in an ERP waveform that typically peaks over centroparietal regions between 300–500 ms after word onset (Kutas & Federmeier, 2011).

Contemporary theorists generally interpret the N400 as an index of the relative difficulty in accessing the lexico-semantic features associated with a word (Federmeier, 2007, 2022; Kuperberg, 2007; Kuperberg et al., 2020; Kutas et al., 2014). There are at least three pieces of evidence to support this interpretation: First, when a word is presented in a plausible context, it evokes smaller N400 responses than when the same word is presented in isolation (Kutas, 1993). This pattern is largely attributed to comprehenders’ abilities to use contextual information to pre-activate (or predict) representations associated with upcoming words, leading to easier lexico-semantic processing. Second, N400 responses to words at the beginning of sentences are often larger than those to words at the end, as the unfolding context generally makes each new word more predictable than the last (Payne et al., 2015; Van Petten & Kutas, 1990, 1991). Finally, the size of the N400 has an inverse correlation with behavioral measures of predictability (e.g. cloze probabilities) such that the greater a word’s predictability, the smaller the N400 response (Kutas et al., 1984; Kutas & Hillyard, 1984; for a computational review of the N400, see Nour Eddine et al., 2022). Taken together, these findings demonstrate that the N400 is sensitive to a word’s predictability given its context—but how do we know which specific aspects of a word are predicted during comprehension?

To understand the nature of comprehenders’ predictions, researchers often compare the N400 responses to violations that share or do not share certain features with an expected word. For example, Federmeier and Kutas (1999) had participants read English sentences with predictable endings like “Chris moped around all morning when he discovered there was no cream cheese. He complained that without it he couldn’t eat his *bagel*.” Then, they either kept the expected word

(*bagel*) in the sentence or replaced it with a word from the same narrow semantic category (*toast*) or a word from a different one (*cake*). The authors found N400 effects for both violations; the effects, however, were reduced for the same-category word (*toast*) relative to the different-category word (*cake*). Moreover, this N400 reduction for same-category words is not observed in sentences where the ending is less predictable, e.g., “Amy was very anxious about traveling abroad for the first time. She felt surprisingly better, however, when she actually boarded the *plane* | *helicopter* | *gondola*.” Thus, the authors concluded that readers predict the semantic features of an upcoming word when the sentence is predictable, leading to smaller N400s when those predicted semantic features appear in the input.

Linguistic prediction can happen at many levels. In addition to semantic predictions, like those described above, there are also studies that demonstrate form-based predictions, which are particularly relevant to the present study. For example, Kim and Lai (2012) created highly constraining sentences like “She measured the flour so she could bake a *cake*.” Then, they either kept the predictable final word or replaced it with a non-word that either shared orthographic features (*ceke*) or did not (*tont*). Results indicated smaller N400 responses to the form-similar non-words (*ceke*) relative to the form-dissimilar non-words (*tont*)—in fact, the size of the N400 responses for *ceke* and *cake* were not significantly different from one another. Numerous studies have replicated this finding using both real words and non-words (e.g. DeLong et al., 2019, 2021; Ito et al., 2016; Laszlo & Federmeier, 2009), and critically, this pattern of findings has been directly linked to predictive processing, as form-based reductions in N400s are not found in less predictable contexts (Ito et al., 2016).

1.3. Using EEG to assess signers' sensitivity to form features during comprehension

Numerous studies have shown signers' sensitivity to semantic violations in sentence contexts (Capek et al., 2009; Grosvald et al., 2012; Gutierrez et al., 2012; Neville et al., 1997). But as we mentioned above, researchers often explain these effects as reflecting bottom-up perception and integration rather than prediction per se (for discussion, see Cates et al., 2013). The strongest evidence for prediction comes from studies demonstrating the prediction of linguistic form. To the best of our knowledge, there are no EEG studies that investigate form-based prediction in sign languages. Specifically, we do not know whether signers show reduced N400s to form-similar violations in predictable but not unpredictable contexts, paralleling the data patterns found in prior written and spoken language studies. There are, however, various studies that use EEG to assess signers' sensitivity to manipulations of form features (i.e. the *handshape*, the *location*, and the *movement* of a sign) during comprehension. These studies are particularly relevant to the present investigation, as they demonstrate signers' awareness of form similarity between signs and their ability to detect form violations in predictable sentence contexts. Thus, we review those findings below.

A handful of EEG studies have used priming paradigms to show that N400 responses to target signs are reduced when preceded by a form-similar sign relative to an unrelated sign (Meade et al., 2018, 2022). For example, the ASL signs for SUMMER and UGLY share *handshape* and *movement* features but differ in their *location*—SUMMER is articulated across the forehead, whereas UGLY is articulated across the face. In contrast, SUMMER and MEDICINE do not share any features. Thus, SUMMER evokes smaller N400s when preceded by UGLY than when preceded by MEDICINE (see Meade et al., 2018). These priming studies do not demonstrate form-based prediction per se,

but they do show that signers are sensitive to form manipulations and experience smaller N400 effects when form features are primed by the prior bottom-up context.

There is one EEG study with findings that could be suggestive of form-based prediction in ASL. Gutierrez et al. (2012) presented native deaf ASL signers with videos of sentences that contained a highly predictable sign, e.g., “ALASKA THEIR ANIMAL WORST BIG THAT **BEAR** EXTREME” (English translation, “*The biggest animal in Alaska is a massive bear*”). The authors either kept the expected sign (BEAR) in the sentence or replaced it with one of four violations. These violations either matched or mismatched the expected sign in their semantic and form features. For example, BAT is semantically related to bear and has the same *location* feature [+semantics, +location]; MONKEY is semantically related but shares no form features [+semantics, –location]; BACKPACK is semantically unrelated but has the same *location* [–semantics, +location]; and finally, DOLLAR is a sign with no semantic or form overlap with BEAR [–semantics, –location].

Results indicated robust N400 effects to all violation conditions between 450–600 ms, consistent with prior demonstrations of semantic incongruity effects in ASL (e.g. Capek et al., 2009; Grosvald et al., 2012; Neville et al., 1997). Interestingly, the N400 effect for the [+semantics, +location] violation seemed to be reduced relative to the others. It is tempting to interpret this finding as an effect of semantic and form-based prediction; however, the one-feature overlap and the no overlap violations evoked similarly sized N400s. Thus, these findings are most consistent with the N400 reflecting the bottom-up processing of these signs and the relative difficulty of integrating them into the context (for discussion of these findings, see also Cates et al., 2013).

In sum, the prior literature on sign language processing has not clearly demonstrated semantic or form-based prediction. To do this, a study would need to contrast the processing of violations in both predictable and unpredictable conditions, finding reduced N400s for form-

similar violations in the former contexts but not the latter. In the section below, we outline the design of the present study and discuss how it aims to assess prediction in ASL.

1.4. The present study

The long-term goal of this project is to better understand how prediction unfolds in sign languages. Our immediate goal, however, is to determine whether signers of ASL make lexically specific, form-based predictions in ordinary contexts. To do this, we used a naturalistic comprehension task known as the *Storytime* paradigm (see [redacted]; [redacted]). In this paradigm, people simply comprehend coherent, naturally produced stories for the sole purpose of understanding the narrative. We embed our experimental manipulations within these stories by changing individual spoken words or signs in carefully controlled ways. There are many advantages to the *Storytime* paradigm: it makes psycholinguistic experiments accessible to a wider range of populations and age groups; it promotes attention to contextual information and arguably enhances predictive processing; and it provides evidence about whether findings from minimal, traditional studies generalize to more naturalistic settings.

In the present study, native deaf signers simply watched someone sign an hour-long story in ASL. Throughout the story, the narrator occasionally produced non-signs with varying degrees of resemblance to the original sign from the story. Specifically, we created two types of non-signs: *handshape-changes* and *all-changes*. In the handshape-change (or *hs-change*) condition, the non-sign matched the original sign from the story in location and movement but mismatched in handshape. For example, if the original sign had a handshape with an extended index finger, we changed the extended finger when producing the non-sign but kept the location and movement constant (see Figure 1 below). In the all-change condition, the non-sign differed from the original

sign in handshape, location, and movement. Importantly, these target signs appeared in both predictable and unpredictable contexts, allowing us to attribute our effects to predictive processing (i.e. predictive effects can only emerge when signers can anticipate upcoming signs and their features).



Figure 1: *Examples of the three sign manipulations in our story context.* In these panels, the narrator is producing the target sign TIME (left panel) and the *handshape-change* and *all-change* variations (middle and right panels, respectively). The handshape-change (or HS-change) non-sign shares location and movement with TIME but uses the “I” handshape instead of the “X” handshape. The all-change non-sign shares no form features with TIME, and it was created by producing FALL with a “7” handshape.

We had three main predictions: First, if predictive processing is occurring, we expect to see differences in the N400 responses to non-manipulated, baseline signs depending on their predictability. Specifically, we should see smaller N400 responses to signs that are predictable relative to signs that are unpredictable (see Kutas et al., 1984; Kutas & Hillyard, 1984). Second, if form-based prediction is occurring, we would expect the non-signs with more phonological overlap, i.e., handshape-changes, to elicit smaller N400 responses compared to all-changes, but only in predictable contexts. Third, we also expect to find another ERP component known as the P600 to all our non-signs. These late posterior positivities are evoked by non-words in reading studies (Kim & Lai, 2012; Laszlo & Federmeier, 2009), auditory studies (██████████), and sign language studies (Grosvald et al., 2012). P600s are often larger in high constraint contexts and in contexts with violations that are similar in form to the original word in the sentence

(Ito et al., 2016; Kuperberg et al., 2020; Laszlo & Federmeier, 2009; Ryskin et al., 2021). Thus, while we expect to find P600s for all non-signs, the P600s may be greater for handshape-change non-signs, which have a closer resemblance to the original sign from the story.

2. Method

2.1. Participants

We recruited 24 native deaf ASL signers; however, we excluded three participants from the final analyses following our pre-registered criteria on OSF ([peer review link](#)). Seventeen participants in the final sample were early signers (exposed to ASL before age 4) and four were late signers (*Mean age* = 31 years, *Range* = 19–52 years). Two participants were left-handed. Participants varied in their highest level of education: four participants had obtained high school or vocational degrees; nine held bachelor's degrees; and eight held a master's degree or higher. Seven participants reported current or prior experience using hearing aids and/or cochlear implants. All participants provided consent and received cash payment for their time. We terminated data collection before reaching a final sample of 30 due to external constraints on our funding sources, recruitment efforts, and research staff availability.

2.2. Stimuli

The present study used a naturalistic comprehension task called the *Storytime* paradigm (see [REDACTED]). We presented a coherent story in ASL in which the narrator occasionally produced a non-sign in ASL (see Figure 1). The story was an ASL translation of an abridged version of an English-language children's story called *Mystery of the Turtle Snatcher* by Kyla Steinkraus. This story stimulus has many useful features: it uses simple language; it has

predictable plotlines that are not widely known to the public; and it is accessible to a wider age range, paving the way for future developmental research. Moreover, this story has been used in prior studies on form-based prediction with hearing adults and children ([REDACTED] [REDACTED]). We selected 180 target signs from the translated story to create a 2×3 manipulation of sign type (*baseline*, *handshape-change*, *all-change*) and predictability (*higher cloze*, *lower cloze*). We also made sure that no lexical items were used as targets more than three times in the story to ensure that no manipulations were repeated.

Sign type manipulations. We first identified 180 target signs unmarked by phonological modifications (e.g. without a shift in signing space). We created our handshape-change non-signs by changing only the handshape parameter of our baseline target signs. In sign languages, handshape refers to the unique configuration of the hand as it is used to form a sign. ASL has a language-specific set of commonly used handshapes including some borrowed from the manual alphabet. When selecting new handshapes for a given sign, we were careful to use handshapes from this established inventory. For example, if the citation form of the target sign included a handshape with an extended index finger, we changed the extended finger when producing the non-sign (see the change from the “X” handshape to the “I” handshape in Figure 1). All-change non-signs were created by finding an existing sign that differed from the intended sign in at least its location and movement. Then, we changed its handshape to make a non-sign, ensuring that this new sign obeyed the phonological constraints of ASL but was not an actual sign.

Additional constraints for creating non-signs. We explicitly avoided creating non-signs that closely resembled other signs in ASL (or an allowable alteration of the target sign). It is common for signers to alter the parameters of their signs, especially when engaging in wordplay or using literary or poetic devices. This is particularly relevant as derivational manipulations of

signs are often used when constructing name signs, and in a story with many new, unfamiliar characters, we wanted to ensure that participants were not comprehending our non-signs as potential name signs.

Another constraint to consider when creating our stimuli is whether the sign is one or two-handed. Two-handed signs can be symmetrical such that if both hands of a sign move, either alternating or simultaneous, then the handshape, movement, and the location must be the same. Two-handed asymmetrical signs have a dominant hand which moves and a non-dominant hand that serves as a base; both hands can have the same or different handshapes. We had 118 two-handed target signs in our original stimulus set. Thus, when constructing the non-signs for these two-handed targets, we strived to maintain and/or create two-handed non-signs; however, in seven instances, we needed to drop the non-dominant hand, as the new location and/or movement features did not allow our signer to produce these non-signs naturally.

Predictability manipulations. To ascertain the predictability of our target signs, we conducted two cloze tasks with native deaf ASL signers via Zoom (for cloze tasks, see Taylor, 1953). In the first cloze task ($N = 11$), participants watched the story with no lexical violations. The video paused before a target sign, and the signers were instructed to provide the next sign in the sentence. The second cloze task ($N = 12$) was identical to the first, except the story contained the non-sign manipulations described above. We used this second cloze task to characterize the predictability of our target signs in the exact video stimuli used in the EEG study (i.e. the story with lexical violations). There was a significant positive Spearman's Rank Order correlation between the by-item cloze probability values observed in the two cloze tasks, $r_s(178) = .61$, $p < .001$. But as we expected, in the cloze task with no lexical violations, the target signs were slightly more predictable than those in the other task: For the task without violations, the average cloze

was 38.9% (first half, $M = 39.1\%$; second half, $M = 38.6\%$). For the task with violations, the average cloze was 31.4% (first half, $M = 31.2\%$; second half, $M = 31.5\%$).

Given the similarity in cloze probabilities across tasks, we combined the data when calculating the cloze probabilities for each target sign. The median cloze probability from these overall averages was 32%, which we used as the boundary for separating higher and lower cloze target signs. Higher cloze targets had an average cloze probability of 58% ($SD = 17.2\%$, $Range = 33\text{--}100\%$). Lower cloze targets had an average cloze probability of 12.8% ($SD = 10.9\%$, $Range = 0\text{--}32\%$).

Sign frequency measures. We also characterized the frequency of our target signs using the subjective frequency ratings from ASL-LEX 2.0 (Schyr et al., 2021). These frequency values were obtained from native deaf ASL signers, who used a 7-point scale to indicate how often each sign appeared in everyday conversation (1 = very infrequently, 7 = very frequently). Higher cloze targets had an average frequency of 5.41 ($SD = 1.04$) and lower cloze targets had an average frequency of 4.98 ($SD = 1.05$). Twelve of our target signs, or at least the dialectal variations that our narrator used, did not appear in ASL-LEX 2.0.

2.3. Procedure

Participants watched an hour-long video of a deaf native signer of ASL signing our story stimulus. Video recordings were presented using PsychoPy (Peirce et al., 2019). Participants sat roughly 100 cm from a TV monitor, and they were encouraged to minimize movement and to keep their faces relaxed. Participants were given a break in the middle of the story to rest. We asked comprehension questions during the break and at the end of the story. Participants were fully

debriefed at the end of the study. All of our experimental procedures were approved by the [REDACTED] ([REDACTED]).

2.3.1. EEG recording and pre-processing procedure

Participants wore an electrode cap (actiCAP SnapCap) containing 31 active Ag/AgCl electrodes that were connected to the EEG equipment, Brainvision's actiCHamp Standard 64 System. We attached two mastoid electrodes (TP9 and TP10) directly behind participants' ears. The EEG data were recorded at a sampling rate of 500 Hz using Brainvision's Recorder (BrainVision Recorder, Version 1.23.0001, Brain Products GmbH, Gilching, Germany). On average, electrode impedances were kept below 20 K Ω . During recording, the ground electrode was FPz and the reference electrode was FP1.

2.4. Analysis plan

The pre-registration for this study, as well as all of the data, analysis scripts, and research materials are available on the Open Science Framework ([peer review link](#)). We analyzed our data using the R statistical computing environment (v. 4.2.2, R Core Team, 2022). Data visualizations were created using EEGLAB (Delorme & Makeig, 2004) and ERPLAB (Lopez-Calderon & Luck, 2014) toolboxes in MATLAB (The MathWorks Inc., 2020), and the *ggplot* package in R (v. 3.4.2, Wickham, 2016).

2.4.1. Determining our spatial and temporal regions of interest

To determine our spatial regions of interest (ROIs), we relied on information from the prior literature and our pilot data. For N400 effects, we selected a centroparietal ROI with eight

electrodes: *Cz, C3, C4, CP1, CP2, Pz, P3, P4*. For P600 effects, we selected a parietal ROI with three electrodes: *Pz, P3, P4*. To determine our temporal ROIs, we used a collapsed localizer technique in which all the conditions being compared are collapsed into one grand average waveform, which is then used to determine the ROI (for discussion, see Luck & Gaspelin, 2017). The grand average can be used to determine ROIs in various ways: simple visual inspection, cluster-mass permutation tests on grand averages, or centering ROIs on peak amplitudes (Luck, 2014; Luck & Gaspelin, 2017). We adopted this last approach and centered a 200 ms time window on the most negative peak (for N400 effects) and the most positive peak (for P600 effects) in the grand average waveforms.

Following our pre-registration, we defined temporal ROIs for each cloze condition separately. We took this step because prior work has shown that predictability can influence the timing of ERP effects (Kutas & Federmeier, 2011; Swaab et al., 2012). For example, N400 effects can emerge earlier in predictable contexts relative to unpredictable ones (Brothers et al., 2015).

2.4.2. Linear mixed effects model specifications

All of the linear mixed effects models were implemented using the *lme4* (v. 1.1-31, Bates et al., 2014) and *afex* (v. 1.2-1, Singmann et al., 2023) packages in R (v. 4.2.2, R Core Team, 2022). All pairwise comparisons were implemented using the *emmeans* package (v. 1.8.4-1, Lenth et al., 2021). We fit our models with the maximal random effects structures justified by our data.

All effects with an absolute value of t greater than 2 are considered significant (Gelman & Hill, 2006). We follow this convention due to the uncertainty in the field about how to best calculate the appropriate degrees of freedom in linear mixed effects models (Baayen et al., 2008). For the sake of completeness, however, we also report the p -values as calculated by the *lmerTest*

package (v. 3.1-3, Kuznetsova et al., 2017). Note, in all of our models, both methods of evaluating significance arrived at the same conclusions.

3. Results

For the present study, we pre-registered a set of primary and secondary hypotheses related to the nature of predictive processing in ASL. Below, we report the findings from a set of linear mixed effects models that aimed to test those hypotheses. Note, these analyses slightly differ from those in our pre-registration due to recommendations during an external review of this work (for more details, see our Supplementary Materials on OSF, [peer review link](#)).

First, if signers in our task are predicting upcoming material, we should see smaller N400 responses to baseline signs in more predictable relative to less predictable contexts. This pattern would be consistent with prior findings that the N400 is reduced as a function of the target word's predictability (Kutas et al., 1984; Kutas & Hillyard, 1984). Second, if signers are engaging in form-based prediction, we should see reduced N400s to handshape-change non-signs relative to all-change non-signs—but only in higher cloze contexts. Third, if signers are sensitive to our non-sign manipulations, we should see P600s in response to all violations. In the prior literature, P600s seem to be sensitive to the constraint of a sentence (e.g. Gunter et al., 2000; van de Meerendonk et al., 2010) and the similarity in form between a violation and the intended word (e.g. Ito et al., 2016; Kim & Lai, 2012; Laszlo & Federmeier, 2009; Ryskin et al., 2021; Vissers et al., 2006). Thus, we should expect larger P600s in more constraining contexts, as well as larger P600s to the handshape-change non-signs relative to the all-change non-signs. We investigate each of these predictions below.

3.1. Visualizing the grand average waveforms and topographic maps

Figure 2 presents the grand average waveforms from centroparietal electrodes (*Cz, C3, C4, CP1, CP2, Pz, P3, P4*) for all experimental conditions. To calculate these waveforms, we collapsed across individual items to create six waveforms for each participant. We then further collapsed these participant-level averages to create the grand averages. Figure 3 presents the topographic maps for the isolated effects of our handshape-change and all-change manipulations.

Visual inspection of these figures suggests three main patterns: First, there is evidence that the N400 responses to the baseline signs are smaller in more predictable contexts. This pattern is consistent with the prior work from written and spoken language studies, suggesting that signers are engaging in predictive processing at some level of representation during naturalistic comprehension. Second, there are robust N400 effects for the two non-sign conditions in higher but not lower cloze contexts. Moreover, these N400 effects are similarly sized, suggesting that our signers did not engage in form-based prediction. Finally, there are robust P600 effects for all non-signs, suggesting that signers detected our violations within the larger story context. These P600s appear larger to handshape-change non-signs than to all-change non-signs, as expected. We do not, however, see any evidence for larger P600s in our higher cloze contexts overall.

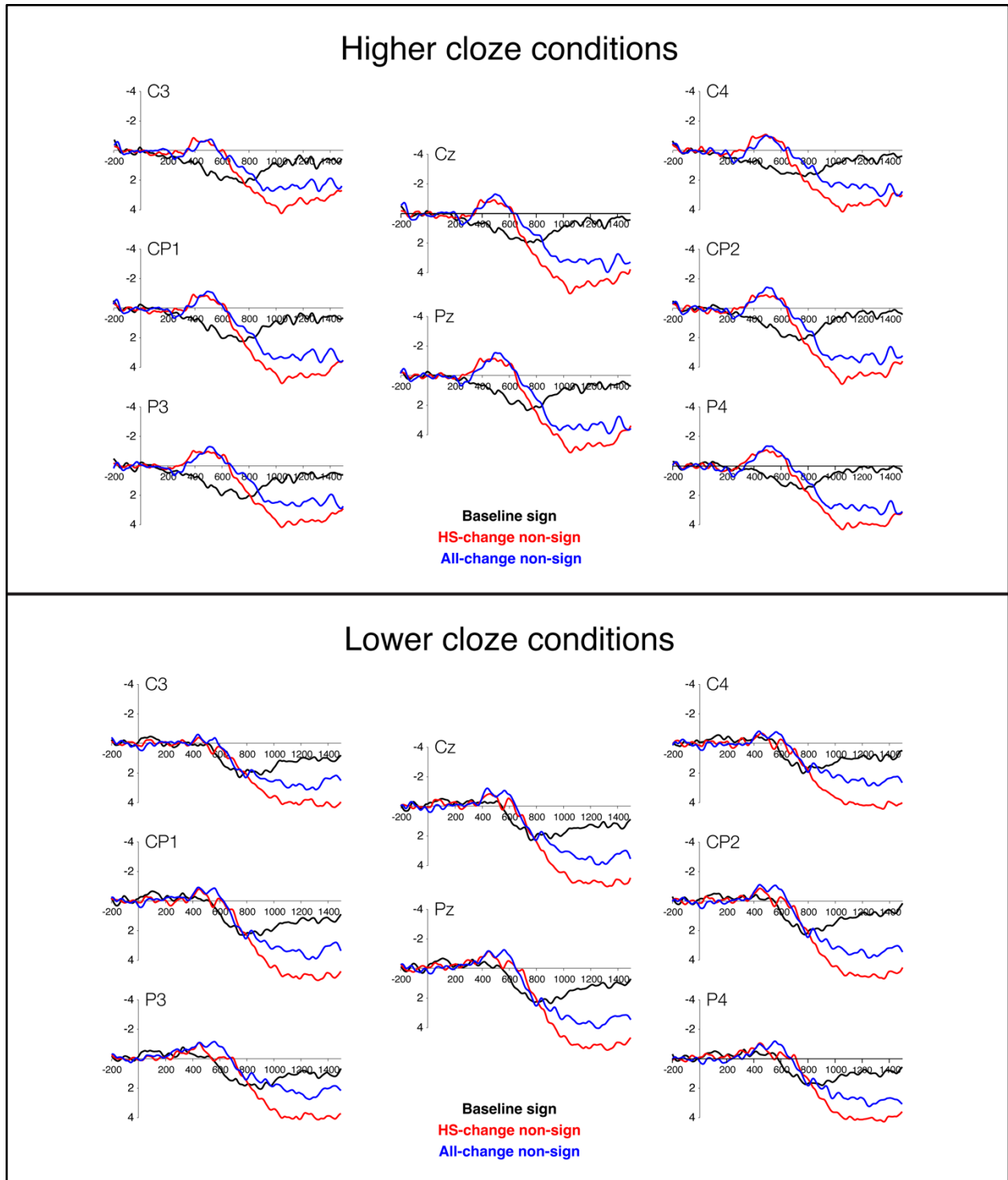


Figure 2: Grand average waveforms for all sign types by predictability. The grand averages (μV) for centroparietal electrodes are presented for higher cloze (top panel) and lower cloze (bottom panel) conditions. The black lines represent the *baseline signs*, while the red and blue lines represent the *handshape-change* and *all-change* non-signs respectively. All waveforms were subjected to an additional low-pass filter of 15 Hz for plotting purposes.

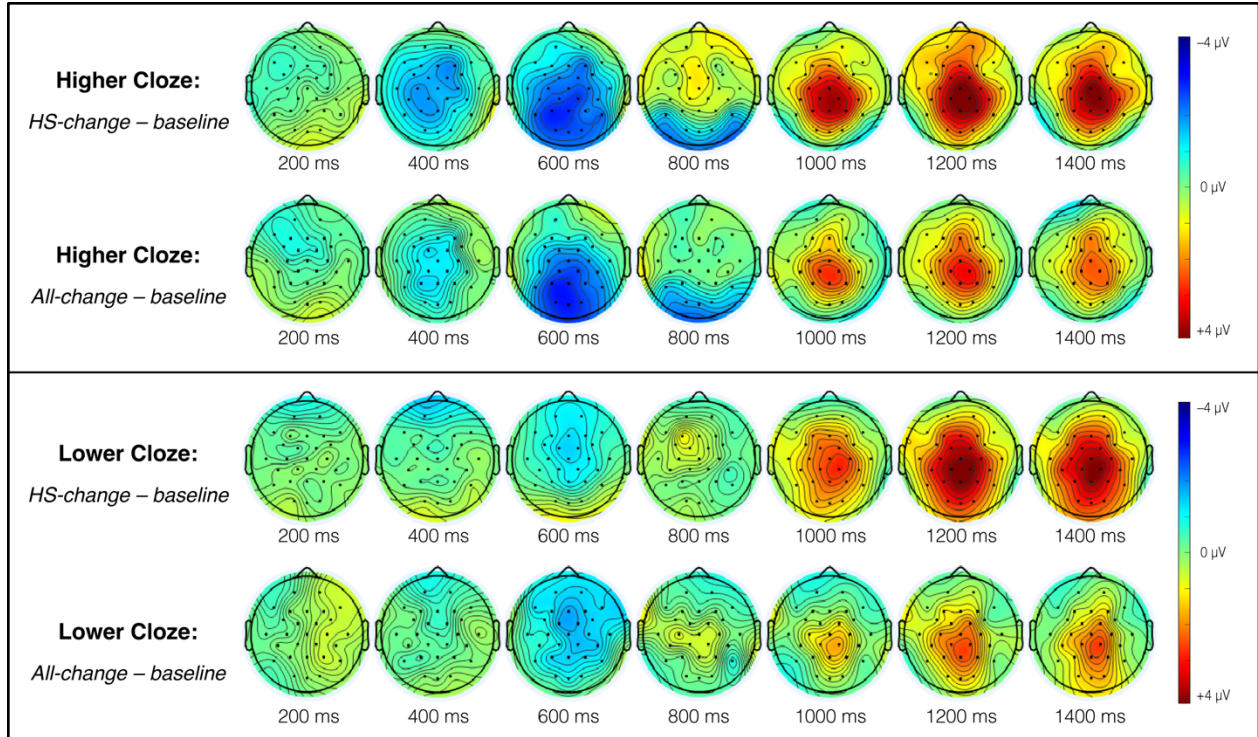


Figure 3. Topographic maps of the ERP effects across predictability. These topographic maps depict the isolated effects of the *handshape-change* and *all-change* non-signs in both higher and lower cloze contexts. These effects are calculated by subtracting the baseline ERP activity from the activity evoked by each violation.

3.2. Do the baseline N400s change as a function of the sign's predictability?

As we discussed above, there appears to be smaller N400 responses to non-manipulated baseline signs in higher relative to lower cloze contexts. In our pre-registration, we planned to assess how our ERP effects changed as a function of sign-level predictability. To do this, we simply used the baseline N400 mean amplitude values from the analysis above, and then implemented a mixed effects model with a single categorical fixed effect of predictability (*higher cloze* = .5, *lower cloze* = -.5). Note, the range of cloze probability values in our items was not well distributed between 0–100%, thus we decided to use a categorical predictor (rather than a continuous predictor) in this analysis. As in prior analyses, we also had a maximal random effects structure for participants and items. Results confirmed that the N400 amplitudes for higher cloze items were

significantly smaller than those for lower cloze items ($b = -1.60$, $SE = .57$, $t = -2.80$, $p = .007$).

Figure 4 shows the average waveforms for higher and lower cloze baseline signs (left panel) and the by-item mean N400 amplitudes by predictability (right panel). This finding confirms that signers were able to pre-activate at least some features of upcoming signs in our task, demonstrating clearly for the first time that prediction occurs during sign comprehension.

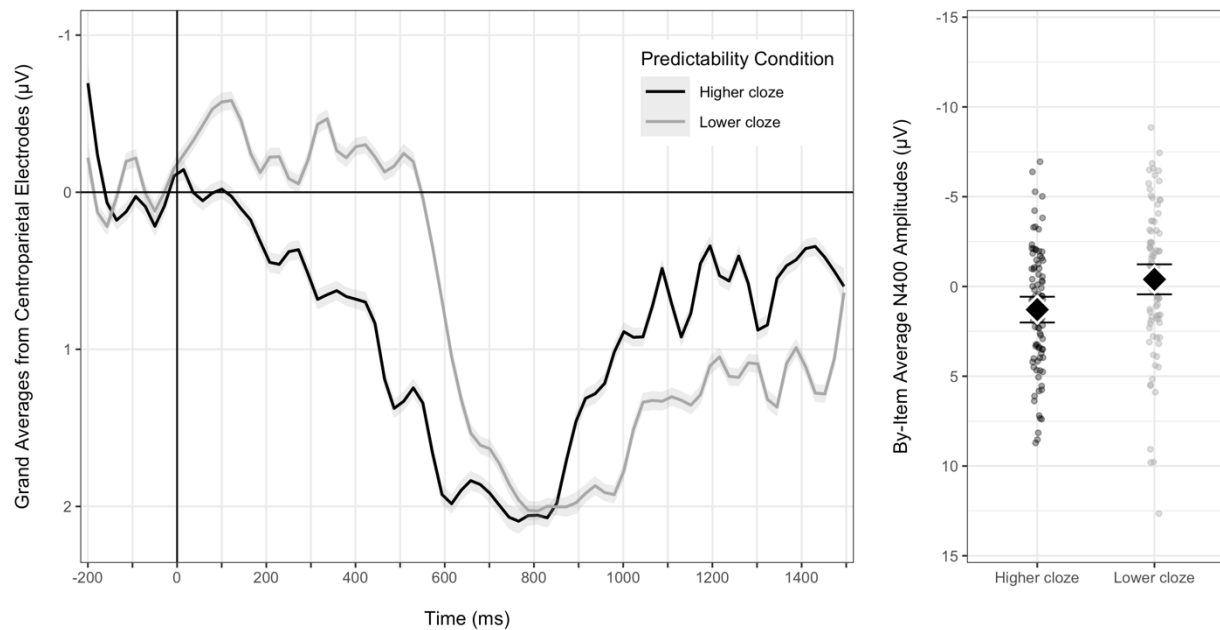


Figure 4: *Average waveforms and N400 mean amplitudes for both baseline conditions.* Grand average waveforms from centroparietal electrodes are plotted for higher and lower cloze baseline signs (left panel, black and gray lines respectively). These waveforms were produced in R and then smoothed using local regression (loess) smoothing techniques. By-item average N400 amplitudes (µV) for all baseline signs are plotted by predictability (right panel).

3.3. Are N400 effects reduced for handshape-change non-signs in higher cloze contexts?

If signers were making *form-based* predictions in our study, we would see smaller N400 responses to handshape-change relative to all-change manipulations—but only when signers were able to predict the original target sign. To test this hypothesis, we calculated mean N400 amplitudes from centroparietal electrodes between 425–625 ms and 345–545 ms for higher and lower cloze conditions respectively. These time windows were selected using the localizer technique described in Section 2.4.1. We then modeled these averages using a linear mixed effects model with fixed effects of sign type (*baseline*, *handshape-change*, *all-change*), predictability (*higher cloze*, *lower cloze*), and their interaction. For sign type, we tested for successive differences using two contrasts: *baseline* vs. *handshape-change*, and then *handshape-change* vs. *all-change*. For predictability, we simply tested for differences between higher and lower cloze conditions (*higher cloze* = $-.5$, *lower cloze* = $.5$). All relevant pairwise comparisons were tested (and corrected for multiplicity) using the *emmeans* package (Lenth, 2021). Finally, we had random intercepts and maximal slopes for participants and items.

Results indicated a significant main effect of sign type for the contrast between *baseline* and *handshape-change* conditions ($b = -1.25$, $SE = .41$, $t = -3.00$, $p = .005$). There was no main effect of sign type for the contrast between non-signs ($b = -0.09$, $SE = .43$, $t = -0.20$, $p = .84$), confirming that handshape-change and all-change non-signs evoked similarly sized N400s. There was also no main effect of predictability ($b = -0.35$, $SE = .39$, $t = -0.91$, $p = .37$); however, there was a significant interaction between sign type and predictability, but only for the contrast between *baseline* and *handshape-change* conditions ($b = 1.94$, $SE = .90$, $t = 2.16$, $p = .039$). Planned pairwise comparisons revealed that this interaction was driven by the presence of N400 effects in higher cloze but not lower cloze contexts: in higher cloze contexts, *baseline* and *handshape-*

change conditions significantly differed ($b = 2.21$, $SE = .59$, $t = 3.75$, Tukey-adjusted $p = .003$) as did the *baseline* and *all-change* conditions ($b = 2.22$, $SE = .60$, $t = 3.71$, Tukey-adjusted $p = .004$). In lower cloze contexts, there were no significant differences in any of the pairwise analyses. Critically, this pattern of results confirms what the Figures show: the N400 effects were larger in higher cloze contexts, but there was no evidence that the N400s were smaller for the violations that were intended to be more similar in form.

3.4. Are the P600 effects sensitive to the non-signs and predictability manipulations?

Given the prior literature, we were interested in three particular patterns for our P600 effects: First, we expected to find P600s in all non-sign conditions. Second, we expected to see larger P600s to violations that more closely resembled the form of the intended sign. Thus, the P600s to our handshape-change non-signs should have evoked larger P600s than our all-change non-signs. Third, P600s are often larger in high constraint relative to low constraint sentences—so, we might expect larger P600s overall in higher vs. lower cloze contexts.

To address each of these predictions, we conducted a parallel analysis to the one described in Section 3.2. In this analysis, however, we modeled average P600 amplitudes calculated from parietal electrodes and time windows of 960–1160 ms and 1225–1425 ms for higher and lower cloze conditions respectively. As in the prior analysis, these time windows were ascertained using the localizer approach described above. The linear mixed effects model for P600 effects had identical fixed and random effects structures and tested the same set of contrasts as the model for the N400 effects. Results indicated significant main effects for both sign type contrasts, confirming P600 differences between *baseline* and *handshape-change* conditions ($b = 3.78$, $SE = .81$, $t = 4.65$,

$p < .001$) and between *handshape-change* and *all-change* conditions ($b = -1.58$, $SE = .62$, $t = -2.53$, $p = .018$). There was no main effect of predictability nor any significant interactions.

These results provide support for two of our three predictions: First, there are robust P600 effects for all non-signs (regardless of predictability); Second, the P600s to handshape-changes are more positive than those to all-changes, suggesting that signers are sensitive to the similarity between the encountered violation and the intended sign given the context. In contrast, there was no evidence for larger P600s in higher cloze relative to lower cloze conditions. This finding is most likely due to the fact that our higher cloze contexts were not extremely constraining, as the average cloze probability for higher cloze target signs was roughly 60%.

4. Discussion

In this study, we found that the N400 responses to predictable signs were smaller than those to unpredictable signs, constituting the first evidence for predictive processing in deaf ASL signers during naturalistic comprehension. We did not find, however, that the N400s to form-similar non-signs were reduced relative to form-dissimilar non-signs when they replaced predictable signs in our ASL story. It is tempting to interpret these findings as evidence that semantic prediction—but not form-based prediction—occurs during sign language comprehension. This conclusion would be quite surprising, as prior work has demonstrated that form-based prediction is readily observed in both written (e.g. DeLong et al., 2021) and spoken language comprehension (██████████). In the remainder of this Discussion, we outline two hypotheses that might explain why form-based prediction is less likely to appear in ASL (Section 4.1). There are, however, a few limitations to the present study, which make us hesitant to accept these conclusions without additional research. Thus, we explore these limitations in Sections 4.2–4.4. Critically, in a new

exploratory analysis, we present preliminary evidence that handshape itself may be the central form feature that is predicted in ASL, resulting in decreased N400 responses for non-signs with more similar handshapes to the originally predicted word (Section 4.4). Finally, we conclude by discussing a few ways in which future research should assess prediction during sign language comprehension (Section 5).

4.1. Two hypotheses about the emergence of form-based prediction in spoken and sign languages

In the Introduction, we described a recent hypothesis from Huettig and Pickering (2019), which proposed that learning to read enhances comprehenders' abilities to predict upcoming words in both written *and* spoken modalities. On this account, learning to read provides the optimal conditions for learning associations between words and broader linguistic contexts (i.e. predicting which words are likely to appear next in a sentence), as well as associations between lexical items and their specific orthographic forms in a language (i.e. predicting what the upcoming word will look like on the page). The authors argue that form-based prediction emerges first in reading because 1) reading is a self-paced process, and there are natural motivations to learn to read faster, and 2) written text is largely invariant across different contexts, making it easy to learn the specific orthographic shapes associated with a particular lexical item. Then, once these predictive skills are established in reading, they begin to emerge in spoken language, acting on the same associations that were strengthened by exposure to the language via text.

As we discussed in the Introduction, this hypothesis states that form-based prediction emerges in spoken language because of the shared systems of semantic, syntactic, and lexical representations. Thus, there is a tacit assumption that predictive skills should *not* transfer from reading to another linguistic modality (or language, more broadly) if it does not share representational systems. In the present study, we tested this hypothesis by looking at the

predictive skills in deaf signers with considerable exposure to reading and writing in English. At face value, our findings are consistent with the literacy hypothesis, as we did not find strong evidence for prediction of form in our ASL comprehension task. To provide definitive support for this prediction, however, we would need to demonstrate that deaf signers actually make form-based predictions when reading in English (see future directions in Section 5).

There is another hypothesis, however, that might account for predictive differences across modalities *without* invoking literacy—namely, sign languages may be less predictable than written and spoken languages in general, resulting in weaker expectations about which signs are likely to appear next. It is well-established that sign languages have access to a wide range of linguistic and depictive devices: *classifier predicates*, *constructed actions*, *lexical signs*, and *fingerspelling* (e.g. Beal et al., 2021; Cormier et al., 2013; Goldin-Meadow & Brentari, 2017; Keane & Brentari, 2015; Padden, 1998). Moreover, sign languages also enjoy freer word orders than the most commonly studied spoken languages (Cheng & Mayberry, 2019; Lillo-Martin, 1986; Napoli & Sutton-Spence, 2014). For example, although ASL follows the same default SVO order as English, signers have access to various non-canonical word orders (under certain pragmatic conditions) like sentence-final pronouns (VS), pre-verbal objects (OV), “verb sandwiches” (VOV), as well as options for null arguments (for examples, see Cheng & Mayberry, 2019).

Taken together, this inherent diversity and flexibility in structuring utterances may increase the number of possible continuations relative to a spoken language that has fewer of these devices. For example, when telling stories, native deaf signers often use constructed actions and classifiers (which can be idiosyncratic and variable) to convey their message in addition to producing individual lexicalized signs (Beal et al., 2021). Thus, prediction of specific signs may be more challenging (and perhaps less useful) during sign language comprehension. Note, this hypothesis

does not imply that sign languages are less structured in the ways that these depictive devices are integrated into the language (for examples of the structured ways that these devices are integrated into sign languages, see Ferrara & Hodge, 2018; Goldin-Meadow & Brentari, 2017). We are simply suggesting that there may be less agreement about how a particular utterance is *likely* to continue within a group of signers relative to a group of hearing individuals, reducing our ability to observe form-based prediction in this population.

To assess this hypothesis, we evaluated the variability in the responses that deaf signers provided in our original cloze tasks. In this task, participants watched our story stimulus and occasionally guessed which sign would come next. All of our target signs were lexicalized nouns in ASL. If deaf signers expected the story to continue as the narrator intended, then they should also provide a noun (regardless of it being the exact noun from the story). Alternatively, if there is greater variability in how signers expect utterances to continue, we may see responses from a wider range of categories (e.g. verbs, adjectives, constructed actions, classifiers). In addition, we compared signers' responses to another set of responses from a parallel cloze task conducted in spoken English with hearing adults and children. Specifically, [REDACTED] had hearing adults and children (aged 5–6 years) watch a cartoon narrative of the same children's story from the present study and guess a subset of target nouns. Thus, these tasks were similar in asking participants to guess target words that were common nouns from the same narrative. What differed between tasks were the set of specific nouns that served as the targets and how the story was presented. For these reasons, the exploratory analysis below focuses on the nature of deaf and hearing participants' incorrect responses rather than how often participants predicted specific lexical items.

Across our three populations, we first selected all incorrect responses (i.e. instances when the participants did not correctly guess the intended word). Then, to allow for cleaner comparisons across populations, we grouped participants' incorrect responses into four syntactic categories: Nouns, Adjectives, Verbs, and Other. The Other category contained responses like adverbs, prepositions, determiners, number words, etc. For signers, this category also contained classifier predicates, constructed actions, and fingerspelling. Again, in all tasks, the target words were all common (lexicalized) nouns. As expected, we found greater variability in the type of responses provided by deaf adults relative to hearing adults and children. Specifically, the proportion of nouns given by deaf adults was lower than the other two groups. Signers appeared to entertain a greater proportion of verbs (20%), adjectives (10%), and other continuations (e.g. *fingerspelling* = 5%, *classifiers* = 4%). In contrast, hearing adults and children rarely provided verbs (adults = 3%, children = 6%), adjectives (adults = 4%; children = 5%), or other continuations from non-noun syntactic categories (adults = 6%; children = 8%).

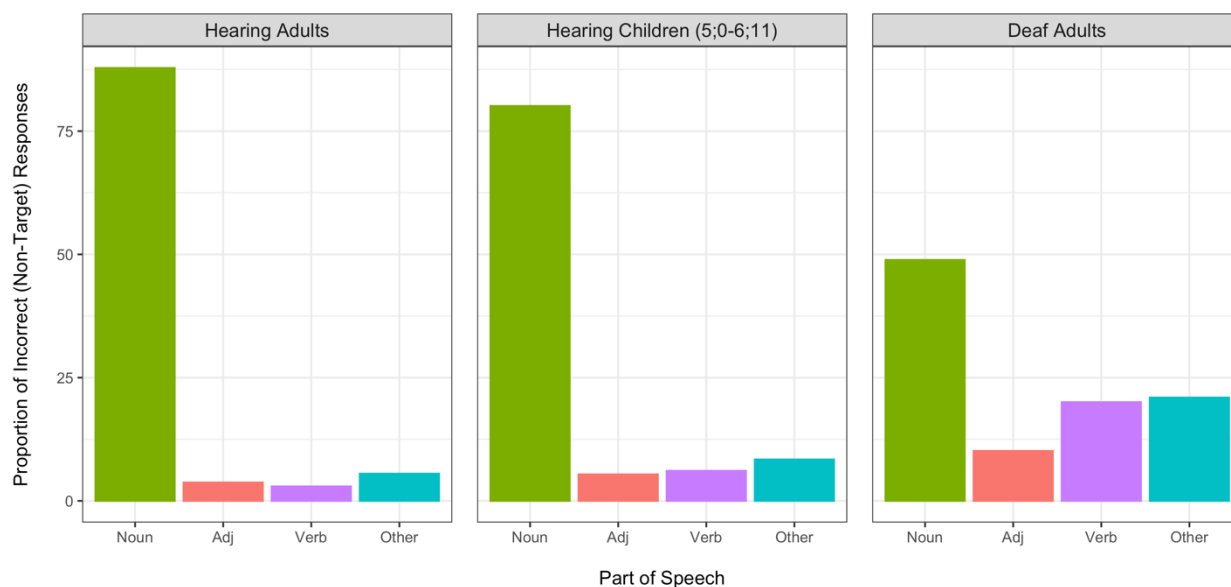


Figure 5. Comparing the variability in cloze responses across hearing and signing populations. We compared incorrect responses provided by hearing and deaf participants in a set of parallel cloze tasks. Specifically, we grouped incorrect responses into four syntactic categories: *nouns*, *adjectives*, *verbs*, and *other*. The *other* category contained responses like *adverbs*, *prepositions*, *number words*, etc. For the

signing population, the *other* category also contained *classifier predicates*, *constructed actions*, and *fingerspelling*. The three panels represent the different participant populations: hearing adults (left), hearing children (middle), and deaf adults (right). Each bar represents the proportion of all incorrect responses (by population) that were classified as belonging to each syntactic category group. The target words in all tasks were common (lexicalized) nouns.

These results tentatively suggest that the baseline predictability of utterances in ASL may be lower due to more variable expectations about how an utterance is likely to continue. For these reasons, we might expect signers to be less likely to engage in form-based prediction during comprehension. At face value, the findings from the present study appear to support this claim. As we mentioned above, however, we are hesitant to draw such a strong conclusion on the basis of a single study. In the final sections below, we discuss a few limitations to the present study and how they could contribute to the observed data patterns.

4.2. Addressing the English-based source material for our story in ASL

A possible concern about the present study is that the source material for our ASL story came from a children's story that was originally written in English. Translations can vary along many dimensions. Some translations are tightly constrained by the original text, with each word or morpheme being re-coded and then presented in as close to the same manner as the target language will allow. A word-by-word translation from English to ASL would likely result in a story that would seem awkward to a fluent ASL user and would lack many of the essential discourse features of an ASL narrative. Thus, one might have concerns that, in the present study, the lack of form-based prediction (and the relative drop in lexical predictability) simply reflects our use of a story that was translated from English to ASL rather than reflecting stable properties of ASL (or ASL users). There are good reasons, however, to believe that this is not the case.

The translation of the original English story was completed by two signers on our team: one native deaf signer and one hearing sign language interpreter. The translation process occurred in multiple steps, which ensured that 1) we were able to retell the story in a way that conveyed the same meaning and intent, and 2) we were taking into consideration any linguistic and cultural differences. To do this, the translators first became familiar with the source text, marking key elements of the plot and aspects of the story that needed cultural modifications (e.g. using noise to gain attention of characters, use of a telephone) or complete removal (e.g. a play on words in English, alliteration, rhyming). There were also aspects of the story that involved humor that did not translate to a retelling of the story in ASL, and so these moments were replaced with culturally appropriate content. Next, the translators divided the English text into parts based on key scenes and translated the story at the paragraph level, using a written gloss system for depicting ASL (including non-manual elements). The translators iteratively revised these ASL glosses, striving for linguistic accuracy (both grammatically and culturally), as well as literary artistry in their final product. Finally, these glosses were produced and recorded by the same native deaf signer, and then further refined after re-watching the naturally produced videos.

We note that these efforts do not completely address the concerns and disadvantages of using an ASL translation of an English story—but we believe that this study took proper precautions to avoid these pitfalls and provide our deaf participants with a highly naturalistic experience. Indeed, many of our participants reported after the study that the storyteller was clearly a deaf signer and that the story was enjoyable and humorous. For our future work, we plan to create a series of stories that are produced off-the-cuff by a native deaf signer, and thus do not rely on source material from another language.

4.3. Addressing the moderate predictability of our target signs

Another methodological limitation in the present study is the relatively low number of highly predictable target signs in our ASL story. To the best of our knowledge, all of the prior studies showing robust form-based prediction used high cloze items with an average cloze probability of at least 80% (DeLong et al., 2021; Ito et al., 2016; Kim & Lai, 2012; Laszlo & Federmeier, 2009). The items in our higher cloze group had an average cloze probability of only 58% ($SD = 17.2\%$, $Range = 33\text{--}100\%$), and only 15 target signs had a cloze probability of 80% or higher. Thus, it is possible that we did not have enough highly predictable target words to properly observe form-based prediction in our signers.

This limitation is something that researchers must address when using naturalistic stimuli. Prior work using cloze methods has shown that highly predictable target words are rather rare in natural discourses (Lowder et al., 2018; Luke & Christianson, 2016; Smith & Levy, 2013). To address this limitation, we would have needed to directly manipulate the plotlines and/or sentence structures in our story to create a set of target signs with artificially high levels of predictability. In doing so, we would have sacrificed the natural variability in word choice and syntactic structure, and perhaps made the story harder to follow for native ASL signers. Thus, we opted to use a fully natural stimulus and then select the highest and lowest cloze signs that occurred in our story.

It is also possible that naturally produced signs (i.e. those produced in conversation between two deaf signers) are generally more predictable than those in our translated story. If that is true, then the moderate predictability of our target signs is a unique property of the particular stimulus that we used. On the other hand, this finding could reflect a meaningful cross-linguistic difference in predictability between written, spoken, and sign languages. Future work should

further investigate the degree to which signers can make explicit lexical predictions using a wider range of naturalistic stimuli and cloze tasks.

4.4. Addressing the variability in handshape similarity across our non-sign manipulations.

Another possibility is that we failed to see the expected form-based modulation on the N400 to handshape-change non-signs because handshapes themselves are such a core feature of lexical identity in ASL. If true, a non-sign with a radically different handshape to the original sign (despite overlaps in location or movement) may not be similar enough to show predictive facilitation. In spoken language studies, for example, predictive facilitation is often only observed on form-similar non-words that share the same phonological onset. In both cases, this reflects our beliefs about which lexical features are most critical for both language processing and linguistic prediction in that modality. On this hypothesis, we might expect that facilitation due to form overlap in ASL will depend on the degree to which the *handshape* of the non-sign is similar to the handshape of the originally predicted sign.

Handshapes naturally vary in their perceptual similarity to one another. In a sign language like ASL, there is a limited set of permissible handshapes (for discussion, see Whitworth, 2011). Moreover, these handshapes—much like phonemes in spoken language—can be further categorized using a set of smaller articulatory and perceptual features (Brentari, 2011; Stungis, 1981). For example, the feature COMPACT describes whether any of the middle fingers are extended: the “A” and “E” handshapes are considered to be [+COMPACT], whereas the “3” handshape is [−COMPACT] (e.g. Lane et al., 1976). In this example, it is clear that the “A” and “E” handshapes could be perceived as being more similar to one another than the “A” and “3” handshapes. Thus, it is possible that some of our non-signs were more similar to the intended target

sign than others. If this is true, and signers are engaging in form-based prediction, we might expect that non-signs with more perceptual (or featural) similarity will show smaller N400s than those with less similarity—but only in the higher cloze items.

There is some EEG evidence in the spoken language literature to support this hypothesis. In a study by Liu et al. (2006), a group of native Mandarin speakers listened to a set of sentences with predictable and unpredictable endings. The authors manipulated sentence-final words to be either the intended word from the sentence or one of the following non-words: a *minimal-onset-mismatch* (a non-word with an onset that mismatched the original by one or two articulatory features); a *maximal-onset-mismatch* (a non-word with an onset that mismatched the original by two or more articulatory features); and a *first-syllable-mismatch* (a non-word with a completely different first syllable than the original word). The authors found N400 effects to all non-words (regardless of predictability). In predictable sentences, however, there were graded N400 effects that varied as a function of the non-words' similarity to the original word: *minimal-onset-mismatch* < *maximal-onset-mismatch* < *first-syllable-mismatch*. In unpredictable contexts, there were no graded N400 effects, as all three non-words elicited similarly sized N400s. These findings provide evidence that listeners are sensitive to these sub-phonemic features, demonstrating graded sensitivity to words' similarities on these dimensions.

To investigate this phenomenon in the present study, we characterized the similarity between our target handshapes and the handshapes used in our non-sign manipulations using a set of similarity values from Stungis (1981). In this prior study, deaf ASL signers identified a series of 20 handshapes in visual noise. The author obtained roughly 400 ratings for each handshape and used those ratings to calculate a confusion matrix (i.e. when given a particular handshape in visual noise, how often did the signer identify it as another handshape). For example, signers incorrectly

identified the “A” handshape as being the “E” handshape in 61/404 trials. Thus, for the purpose of this exploratory analysis, “A” and “E” would have a similarity value of 61. In contrast, “A” and “3” were only confused once, giving them a similarity value of 1.

Unfortunately, Stungis (1981) only tested a set of 20 handshapes—thus, we needed to restrict our analyses to the subset of trials in which the target and *both* non-sign handshapes appeared in that original study. Initial characterization of our non-sign conditions revealed comparable degrees of similarity between their handshapes and those of their respective baseline target signs (Handshape-change non-signs: *Mean similarity* = 10.6, *Range* = 0–37; All-change non-signs: *Mean similarity* = 10.4, *Range* = 0–36). We then split the items with similarity values greater than 7 (the median value) into a *Similar* group and all others into a *Dissimilar* group. We also separated items into the higher and lower cloze groups from our prior analyses.

Following the logic above, we might expect that the N400s to non-signs with highly similar handshapes would be smaller than the N400s to non-signs with less similar handshapes in higher but not lower cloze conditions. In Figure 6, we present the waveforms from the subset of items in which one non-sign condition is categorically more similar in handshape to the intended target than the other condition (based on the median split). Specifically, we show that more similar non-signs (regardless of originally being handshape-change or all-change) evoked reduced N400 responses relative to dissimilar non-signs—but only in higher cloze conditions. In lower cloze conditions, there is no difference between the two non-sign manipulations (regardless of their handshape similarity).

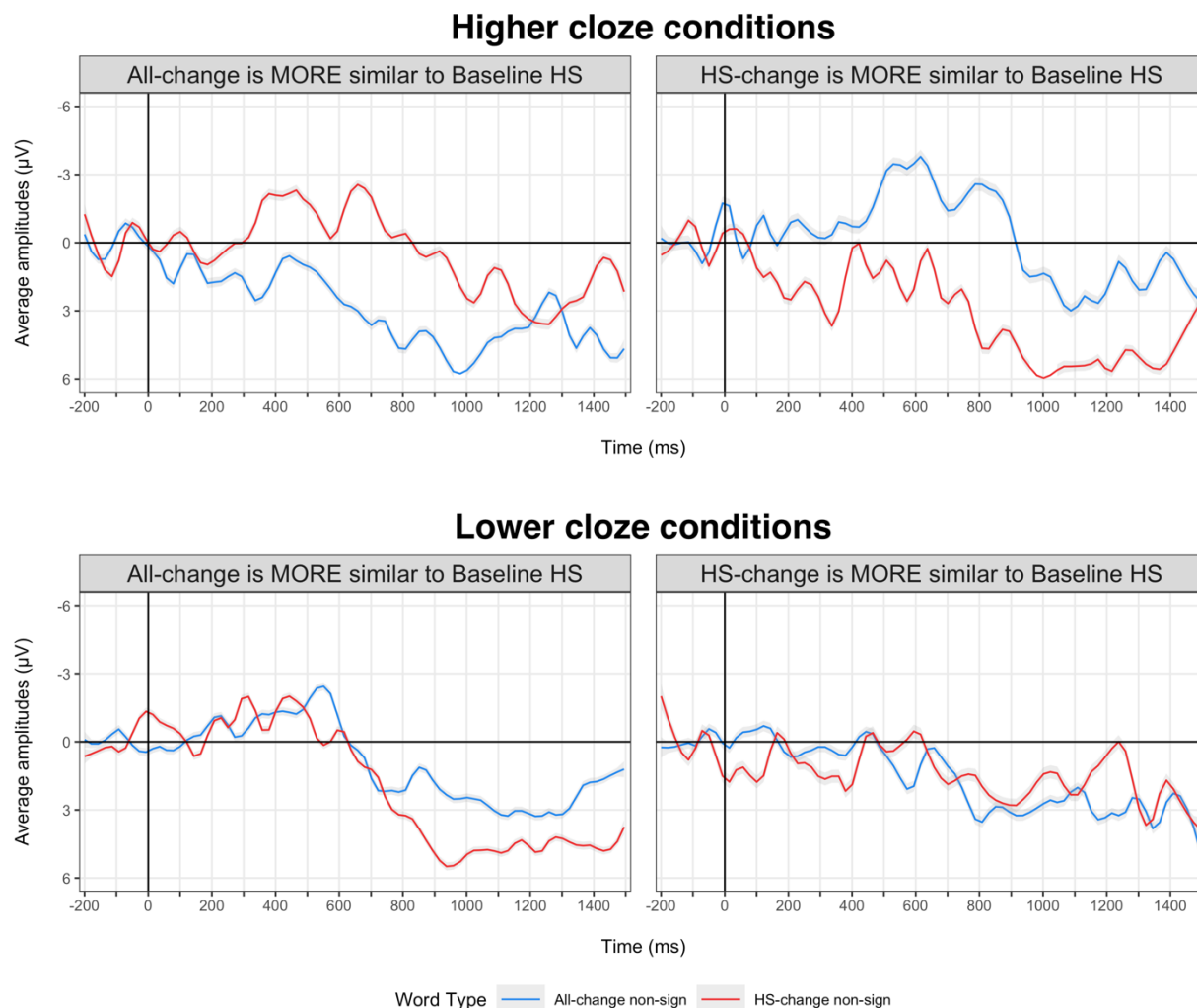


Figure 6. Average waveforms for non-signs across handshape similarity and predictability. Grand average waveforms from centroparietal electrodes are plotted for higher cloze (top panels) and lower cloze conditions (bottom panels). The waveforms from trials in which the all-change non-signs had a more similar handshape to the target than the handshape-change non-signs are on the left. The waveforms from trials with the opposite relationship are on the right. The red lines indicate handshape-change non-sign average waveforms, whereas the blue lines indicate all-change non-sign average waveforms. All waveforms were produced in R and then smoothed using local regression (loess) smoothing techniques.

One limitation to this analysis, however, is that there were only 36 items in which 1) the handshapes from *both* non-signs and the target sign appeared in Stungis (1981), and 2) the two non-signs categorically differed in their handshape similarity (i.e. one was classified as *Similar*, and the other as *Dissimilar*). Thus, in order to draw any strong conclusions from these patterns, we would need additional data. In an effort to gain more power to observe an effect, we simply

grouped non-signs by the similarity of their handshapes to the target words that they were meant to replace (ignoring whether they were originally handshape-change or all-change non-signs). By collapsing in this way, we were able to use all of the data from trials with similarity values for at least one non-sign, increasing our number of usable items to 103. In Figure 7, we show the grand average waveforms by predictability for similar and dissimilar handshape manipulations relative to their baseline controls (left panel). Figure 7 also presents the mean N400 amplitudes from our previous analysis, but rather than separating them by sign type, we separated these values by similarity group and predictability (right panel). Note, the original sign type for each observation is depicted as a red point (handshape-change non-sign) or a blue point (all-change non-sign).

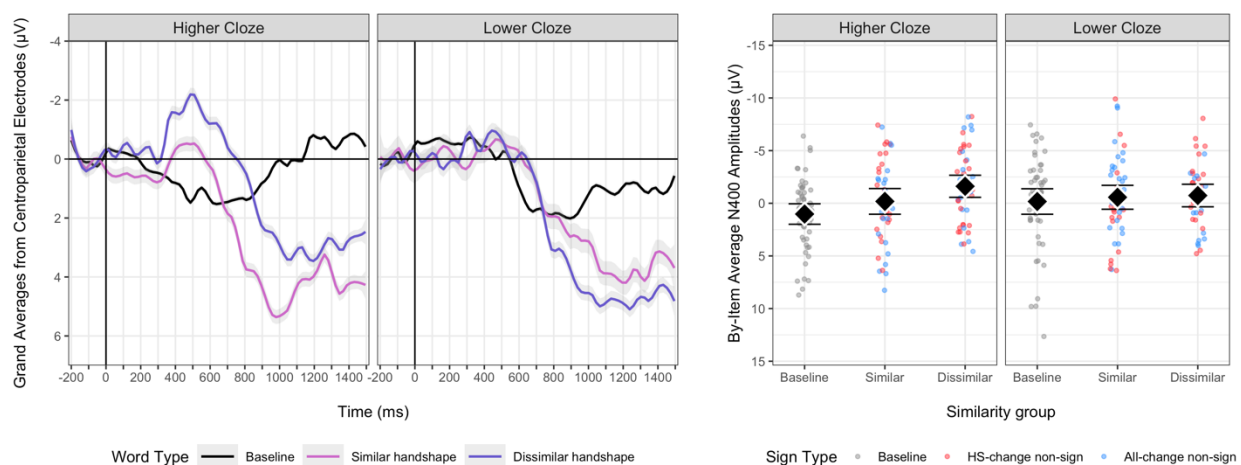


Figure 7. Average waveforms across handshape similarity groups and predictability. Grand average waveforms from centroparietal electrodes are plotted for higher cloze and lower cloze conditions. These waveforms indicate when the non-signs had a similar handshape to the target (pink lines, left panels) or a dissimilar handshape (purple lines, left panels). Item-level mean N400 amplitudes for baseline signs, similar non-signs, and dissimilar non-signs are plotted in the right panels. Gray points indicate baseline N400s, whereas red and blue points indicate the original non-sign type (e.g. handshape-change or all-change non-signs, respectively). Black diamonds represent the mean amplitude for the group. These waveforms were produced in R and then smoothed using local regression (loess) smoothing techniques.

These exploratory observations provide tentative evidence for form-based prediction in our study. The N400 responses to violations with similar handshapes to their respective target signs were reduced in predictable but not unpredictable contexts. Interestingly, this reduction did not

appear to be sensitive to whether the non-sign matched in location and movement with the target sign. Both handshape-change and all-change violations produced smaller N400s when their handshapes were perceptually similar to (and easily confusable with) the target handshape (Figures 6 and 7).

These data suggest that signers are particularly sensitive to handshape features during comprehension. Recent evidence from the priming literature provides tentative support for this claim. In a study by Meade et al. (2022), native deaf ASL signers showed equally robust priming effects when the target shared the same handshape with the prime, and when the target shared both the handshape *and* location with the prime. Thus, handshape could be a particularly prominent feature of sign recognition, perhaps similar to consonants in spoken language. Like consonants, handshapes are more perceptually distinct, greater in number, and are less likely to have a grammatical component. For example, altering a handshape is less likely to change the identity of that sign relative to altering its location or movement. Another possibility is that signers only generate expectations about handshapes, and not about location or movement, during comprehension. Handshape features seem to be fully articulated well before location or movement—thus, it may be advantageous to predict early emerging parameters relative to late emerging parameters. We hope that future work will explore whether similarities in handshape alone are enough to explain the modulation of the N400 during sign comprehension.

5. Conclusion and future directions

The present study finds clear evidence for predictive processing in ASL signers during a naturalistic comprehension task. The N400 responses to non-manipulated baseline signs were smaller as a function of the signs' predictability in context. In most contemporary theories of the

N400, this evidence is argued to reflect the pre-activation of lexico-semantic features associated with a word (e.g. Federmeier, 2007; Kuperberg, 2007; Kuperberg & Jaeger, 2016, p. 39; Kutas & Federmeier, 2011). We also explored the degree to which handshape similarity could have impacted our findings. In a series of exploratory analyses, we found that N400s were reduced to violations that have similar handshapes to the target sign that they were meant to replace in the story. These reductions, however, only seemed to emerge in predictable contexts, presumably when signers were able to pre-activate the handshape of an upcoming sign. Moreover, these reductions seemed to be independent of whether the violation overlapped in location and movement features with the intended sign, generating open questions about which features are actually predicted during naturalistic sign comprehension. In sum, the present study sought to characterize prediction in sign language, and we found promising (but tentative) evidence that prediction does occur in ASL—but confirmation of these conclusions will require further theoretical and statistical testing.

In our future research, we plan to continue investigating predictive processing in ASL. Specifically, we have identified three projects to pursue: First, we plan to replicate the finding that the N400 response to non-manipulated signs has an inverse linear correlation with the predictability of that sign in its context (e.g. DeLong et al., 2005; Kutas & Hillyard, 1984). In the present study, our target items were not evenly distributed across the possible range of cloze probabilities, restricting our ability to assess any continuous relationship between the N400 and the predictability of a sign. We therefore plan to identify and use a stimulus with target items that sufficiently represent the continuous nature of cloze probabilities.

Second, we hope to further explore signers' sensitivity to handshape features and to determine whether signers specifically target this parameter in their predictions. To do this, we

will need a careful set of stimuli that balances the perceptual similarity between the three main parameters in ASL.

Finally, in the present study, we narrowly focused on how reading experience influenced prediction in one's primary linguistic modality. To do this, we investigated the nature of prediction in signers who are also fluent readers of English. There are, however, interesting questions about whether such signers make robust form-based predictions while reading. Much research in psycholinguistics has shown interesting similarities and differences between skilled deaf and hearing readers: deaf readers process visual word forms more efficiently than hearing readers, and they also have wider perceptual spans (e.g. Traxler et al., 2021). Both skilled deaf and hearing readers show sensitivity to orthographic and semantic information during written language comprehension—but deaf readers seem to rely less on phonological information than hearing readers (see Emmorrey & Lee, 2021; Lee et al., 2022). Given these differences, an important next step in this research program is to understand prediction in written language with highly literate and skilled deaf readers. A simple approach to this question would be to have signers participate in the prior reading studies that clearly demonstrated form-based prediction with hearing participants.

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